Symbolic Music Generation

POP909 | Causal Transformer models trained from random initialization

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Code, own weights and all listening files: https://github.com/alexsu30812-gif/hw2

Task	Required result	Submitted files
Basic a	5 settings x 20 complete 32-bar samples; select t12	20 MIDI + 20 WAV
Basic b	3 prompts x 3 settings; original 8 + 24 new bars	9 MIDI + 9 WAV
Bonus	10 melody-conditioned 16-bar piano accompaniments	10 MIDI + 10 WAV

Beyond the minimum: an equal-token context experiment and a paired melody-conditioning ablation, with uncertainty and concrete register/harmony failures.

Hosting note: GitHub replaces the slides' cloud-drive folder at the user's request; TA acceptance of that substitution is unverified.

Data, representation and model

Data and timing

Only the 896 supplied training songs; exclude all 13 prompt songs. Diagnostic split: 851/45 songs (seed 20261022); final own checkpoints observe all 896. The final fit has no held-out validation score.

Use beat_midi.txt for beat/bar boundaries: source MIDI meters can be wrong. Export n annotation beats as n/4; retain pickup bars.

Causal Transformer

6 layers, width 384, 6 heads; 12,482,688 parameters. Pre-LN, GELU, dropout 0.1, tied embeddings, learned absolute positions (capacity 4096; effective context 2048).

Random initialization; predict the next token with cross-entropy, ignoring padding. Bonus continues our own refined basic weights.

REMI-style event tokens

BOS, [Bar, Meter, Tempo, (Pos, Pitch, Vel, Dur)*], EOS. Vocabulary: 683; 8 subdivisions/beat, 32 velocity bins, durations 1..256 grid steps. Basic merges MELODY/BRIDGE/PIANO; bonus retains MELODY/PIANO roles.

Optimization and decoding

AdamW (betas .9/.95, weight decay .1), norm clip 1; warmup + cosine LR; legal transposition -5..+5. Decode with grammar masks, pitch 21..108, at most 64 notes/bar. Overflow retains recent complete bars and resets positions/KV cache.

Training lineage and the context experiment

New non-padding targets at each stage; optimizer resets between stages. M4 Pro / PyTorch MPS.

Stage	Steps	New targets	Context	Songs	Batch	LR	Val CE
Basic diagnostic	2000	13.751M	2048	851	4	3e-4	1.97121
Basic refined	4000	27.544M	2048	851	4	1.8e-4	1.71925
Context 2048	579	4.007M	2048	851	4	8e-5	1.70637
Context 4096	696	4.003M	4096	851	2	8e-5	1.74084
Basic final	4081	28.006M	2048	896	4	5e-5	not held out
Roles diagnostic	1194	8.003M	2048	851	4	1.5e-4	1.50460
Roles final	1794	12.005M	2048	896	4	5e-5	not held out

Basic: diagnostic -> refined -> context 2048 -> final. Roles: refined -> roles diagnostic -> roles final. Validation CE falls from 2.36666 (step 500) to 1.71925 (refined).

Extra experiment: same refined parent, LR 8e-5, ~4M new targets per branch.

2048: CE 1.70637 in 309.1s; 4096: CE 1.74084 in 591.5s. Both use the same 45-song, 2048-token validation; token budgets differ by 0.086%. Select 2048 before final refitting. One run each; batches/steps differ, the parent favors 2048, and validation uses 2048. This does not establish that longer context is generally worse.

Basic a: all sampling settings and H4 / GS

Same basic_final checkpoint, CPU float32; seed 20261022+i (i=0..19). Unconditional 4/4, 120 BPM; means +/- piece SD.

Set	T / top-p / top-k	H4	GS (official)
Real reference (45)	same grid adapter	2.6398 +/- 0.1225	0.970831 +/- 0.013551
t08	T=0.8; p=0.95; k=0	2.6072 +/- 0.1202	0.997488 +/- 0.002611
t10	T=1; p=0.95; k=0	2.7255 +/- 0.1197	0.992609 +/- 0.004625
t12 [selected]	T=1.2; p=0.95; k=0	2.7846 +/- 0.1132	0.970166 +/- 0.024619
k16	T=1; p=1; k=16	2.7256 +/- 0.0571	0.981920 +/- 0.007164
k32	T=1; p=1; k=32	2.7061 +/- 0.1117	0.982149 +/- 0.009521

H4: pitch-class entropy in 4-bar windows. GS: bar-onset pattern similarity. Reference: one deterministic eligible 32-bar crop per diagnostic song, included in final fitting (descriptive reference); all use the same Pos8-to-Pos16 adapter. All 100 generated pieces pass length/note checks; zero skips.

Selection rule: minimize the mean absolute H4/GS gap in real-reference SD units. t12: 0.6155; k32: 0.6882; k16: 0.7593. This is selection on the reported samples, not an independent test or a musical-quality score.

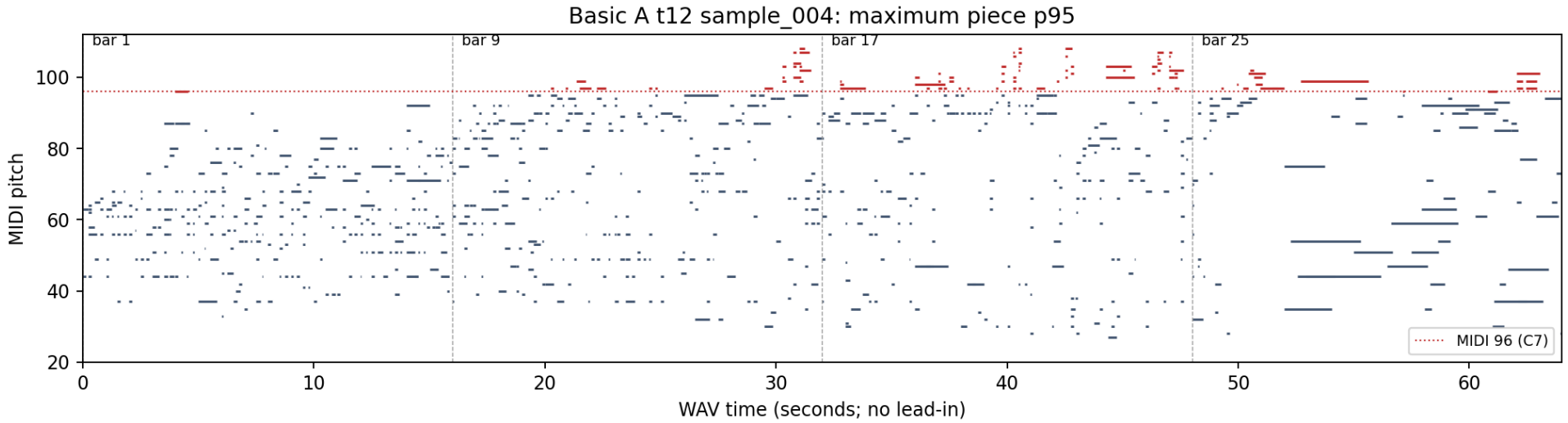
Basic a: similar metrics can hide register drift

Set	Notes/bar	GS active
t08	38.952	0.98995
t12	21.045	0.88066

GS active is native-grid onset similarity, not official GS. t08 is dense with nearly fixed onsets; t12 reduces both (real density: 22.953 notes/bar). Mean piece-p95 pitch is 92.385 vs real 78.363 (+14.022 semitones). H4 discards octave; GS ignores pitch.

004 has the largest piece-p95 (100). Bar 26, 50-52 s: 7 of 9 onsets reach >=96 (C7). This is a descriptive warning, not proof of poor sound.

Selected sample 004 | whole-piece piano roll; dotted C7 line is a register marker, not a correctness cutoff.



Basic b: all nine continuations and best labels

8 supplied bars + 24 generated bars. Restore every original prefix note; all nine preserve pitch/velocity/timing identity within export rounding.

Prompt	Set	T	Boundary score	Density ratio	Pitch jump	Choice
001	t08	0.8	1.0342	1.926	4.546	
001	t10	1	1.0499	1.815	3.733	
001	t12	1.2	0.4552	1.333	-0.296	BEST
002	t08	0.8	0.8008	1.239	1.037	
002	t10	1	0.7137	1.522	2.026	
002	t12	1.2	0.5566	0.913	2.588	BEST
003	t08	0.8	0.4288	1.113	-2.144	
003	t10	1	0.3285	1.113	0.941	BEST
003	t12	1.2	1.0464	1.226	7.108	

All use top-p=.95, top-k=0 and CPU float32; fixed seed per prompt (20261022 + prompt index 0..2). Compare 2 bars before/after the seam: |log density ratio| + |mean pitch jump|/12 + (1 - pitch-class Jaccard); lowest wins within each prompt.

Best labels measure local symbolic continuity. The assignment requires no objective metrics here; no human listening preference is claimed.

Basic b: what changes after the boundary?

Each cell lists bars 9-16 / 17-24 / 25-32, in that order. Saved-token statistics; * = original boundary-rule choice.

Prompt	Set	Notes/bar: early / mid / late	Mean pitch: early / mid / late	Velocity: early / mid / late
001	t08	31.000 / 32.875 / 37.000	63.14 / 62.69 / 66.02	101.73 / 119.52 / 120.96
001	t10	34.875 / 46.625 / 38.750	63.94 / 67.31 / 67.26	105.44 / 109.25 / 103.95
001	t12 *	24.250 / 26.750 / 30.125	64.38 / 68.13 / 64.12	92.47 / 93.53 / 97.78
002	t08	34.625 / 33.875 / 42.875	67.90 / 70.56 / 69.66	85.09 / 85.97 / 85.95
002	t10	38.500 / 35.875 / 32.125	65.05 / 68.14 / 66.09	84.09 / 83.97 / 84.27
002	t12 *	20.250 / 19.125 / 23.875	68.14 / 65.87 / 65.73	92.05 / 84.95 / 84.30
003	t08	29.000 / 35.000 / 38.125	59.03 / 61.47 / 63.84	77.00 / 77.14 / 77.72
003	t10 *	30.625 / 36.000 / 27.250	63.58 / 68.05 / 64.33	76.58 / 74.92 / 75.49
003	t12	33.875 / 27.000 / 9.250	69.08 / 75.30 / 85.27	74.43 / 76.44 / 77.03

Window time boundaries: 001: 30.0/60.0/89.9/119.9s; 002: 17.8/35.6/53.3/71.1s; 003: 27.8/55.7/83.5/111.3s. Prompt bars 1-8: 001: 15.50 notes/bar, pitch 60.94; 002: 22.12 notes/bar, pitch 65.98; 003: 22.75 notes/bar, pitch 60.65.

003/t12: late density falls 72.7% and pitch rises 16.19 semitones relative to bars 9-16; selected t10 stays much closer (30.625 -> 27.250 notes/bar, pitch 63.58 -> 64.33). A smooth seam does not establish later stability. Full per-bar CSV/JSON is included.

Bonus: melody-conditioned accompaniment

Own roles_final: insert each bar's MELODY, then sample PIANO (T1, p=.95, k=0), seeing past bars/current melody. Input: melody.mid + beat_midi.txt; no reference piano/chords. Restore original melody; export MELODY + PIANO.

Prompt	CA	OA_P	OA_V	OA_D	OA_IOI
001	0.1250	0.8628	0.1864	0.7794	0.6378
002	0.0625	0.8356	0.5613	0.6928	0.6695
003	0.1875	0.8577	0.6127	0.6806	0.5059
004	0.0000	0.8572	0.5564	0.5609	0.2987
005	0.1250	0.8577	0.7719	0.7475	0.4433
006	0.1250	0.8464	0.6240	0.7007	0.4136
007	0.0625	0.8775	0.7011	0.8512	0.7225
008	0.1250	0.8549	0.7008	0.7925	0.5936
009	0.2500	0.8423	0.7084	0.7376	0.8253
010	0.0625	0.8599	0.7080	0.8150	0.5772
Mean	0.1125	0.8552	0.6131	0.7358	0.5687

CA: 24-triad agreement with bar annotations. OA_P/V/D/IOI: distribution overlap in pitch class / velocity / duration / inter-onset interval. All 16 bars evaluated per prompt; zero skips.

Extra experiment: remove melody conditioning

Same roles checkpoint, 10 prompts, seed schedule and T=1 / p=.95. Remove melody-note tokens, retain bar/meter/tempo/track headers; restore the complete original melody in both exports. Evaluate official CA/OA; bootstrap 5,000 paired whole-prompt resamples.

Metric	Normal	Empty melody	Difference	95% low	95% high
CA	0.1125	0.0375	0.0750	0.0250	0.1313
OA_P	0.8552	0.8504	0.0048	-0.0034	0.0128
OA_V	0.6131	0.3779	0.2352	0.1359	0.3382
OA_D	0.7358	0.6002	0.1356	0.0343	0.2615
OA_IOI	0.5687	0.4933	0.0754	-0.0976	0.2574

CA rises by 0.0750, OA_V by 0.2352 and OA_D by 0.1356; their intervals stay above zero. OA_P and OA_IOI intervals include zero. Melody access affects measured descriptors, with weaker evidence for pitch-class and onset-distribution overlap.

Limits: only 10 prompts; five unadjusted intervals. Empty melody is out of distribution and changes context length/sample history. This does not isolate perceived quality. The empty condition is evaluation-only and is not submitted.

Harmony failures: which bars disagree?

Official 24-triad estimator vs maximum-duration annotation per bar. Ground-truth piano is evaluated separately, never supplied to generation.

Prompt	Bar (WAV time)	Reference	Generated	GT estimator	Result
004	1 (0.00-3.43s)	Ab:maj	F:min	Ab:maj	mismatch
004	2 (3.43-6.86s)	Eb:maj	C:min	Eb:maj	mismatch
009	1 (0.00-2.76s)	E:maj	Db:min	E:maj	mismatch
009	2 (2.76-5.52s)	B:maj	E:maj	B:maj	mismatch
009	3 (5.52-8.28s)	Db:min	Db:min	Db:min	match
009	4 (8.28-11.03s)	B:maj	B:maj	B:maj	match

- Overall CA: 18/160 bars. 004: 0/16; 009: 4/16 (bars 3, 4, 5, 14). The GT estimator matches 13/16 for both and 0.7625 overall, so it is an imperfect comparison, not a strict upper bound.

- 004 bars 1-2 are estimated as relative minors (Ab:maj -> F:min; Eb:maj -> C:min): each pair shares 2 of 3 triad tones. A CA mismatch need not mean audible dissonance.

- High OA_P can coexist with wrong chord order. Test train-only chord/phrase supervision next; the current outputs and original selections remain unchanged.

Discussion and reproduction

- Training: at this warm start/budget, 2048 context gives lower validation CE than 4096. It costs less here, but batch size and the 2048-trained parent confound a general context-length claim.

- Generation: warmer sampling reduces dense, nearly fixed onsets; H4/GS still hide register drift. Melody conditioning improves CA/velocity/duration descriptors, yet only 18/160 bonus chord labels match. A chord/phrase objective and better long-range memory are useful next experiments.

- Secondary temperature check: the custom equal-weight mean of 5 bonus metrics selects T1 (.5771) over T1.2 (.5726). Paired 95% CI [-.0216, .0295] includes zero; this is in-sample tuning, not proven better music or a course-grade formula.

- Audio limit: all nine WAVs (15 min 25 s) reached an evaluation-only Qwen critic, but it invented music on silence. Its prose is not used as listening evidence. No human listening ratings or official SI experiment are claimed.

- Reproduce: pip install -r requirements.txt, then src/infer.py; exact seeds/settings, output paths and WAV commands are in README. Package: 39 MIDI/WAV pairs + 2 own checkpoints, vocabulary, code, figures and full experiment JSON/CSV. Training data, external critic weights and caches are excluded.

Sources and AI assistance

- Course Homework 2 (Bo-Rui Chen, 2026): task pp.11-18,22-28; submission pp.29-32; grading p.33; AI permission p.34; rules p.35.

- Wang et al. POP909: A Pop-Song Dataset for Music Arrangement Generation. ISMIR 2020. https://arxiv.org/abs/2008.07142

- Huang & Yang. Pop Music Transformer: Beat-based Modeling and Generation of Expressive Pop Piano Compositions. ACM MM 2020. https://arxiv.org/abs/2002.00212

- Ren et al. PopMAG: Pop Music Accompaniment Generation. ACM MM 2020. Inspiration for role-aware accompaniment; our weights are trained from scratch.

- Course HW2_scripts/tokenization.py, eval_basic.py, eval_bonus.py; MusDr side_utils.py: https://github.com/slSeanWU/MusDr/blob/master/musdr/side_utils.py

- Chu et al. Qwen2-Audio Technical Report (2024). https://arxiv.org/abs/2407.10759 ; evaluation-only model: https://huggingface.co/mlx-community/Qwen2-Audio-7B-Instruct-4bit (Apache-2.0).

- Blaizzy/mlx-audio v0.6.1 (MIT): https://github.com/Blaizzy/mlx-audio . Official HF audio frontend substituted after numerical comparison; full model revision/provenance is saved.

AI assistance: Codex assisted implementation, execution, result aggregation, analysis and English drafting. All numerical results come from saved runs. The external Qwen critic is evaluation-only, failed calibration, and is not either submitted composition model. Full source hashes, checkpoint ancestry, per-piece results and package checks remain in results/report_data.json, FILE_SHA256.json and SUBMISSION_CHECKS.json.